Dr. Babasaheb Ambedkar Technological University, Lonere.
B. Tech. Course in Electrical Engineering
Subject: Power System Stability and analysis (EE805)
Sumer examination May 2019

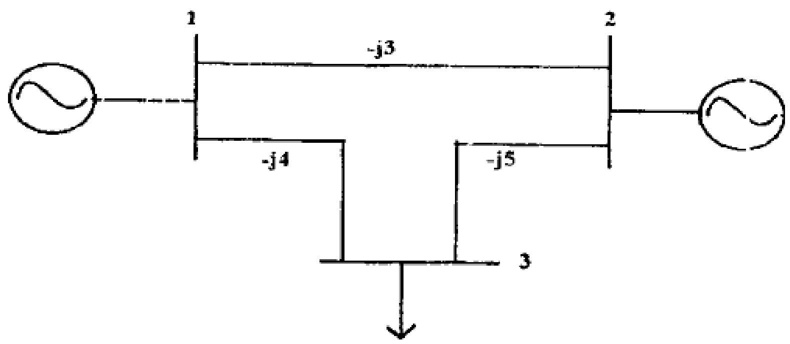
Semester: VIII **Class: B. Tech (EE)**
Date: 24/05/2019 **Time: 3 Hours** **Total Marks: 70**

Instructions to the Students:
1. **Read questions carefully.**
2. **All Question are compulsory.**
3. **If any data is missing is a part of question. Assume suitable data giving reason.**

Q.1
A) Explain slack bus, generator bus and load bus. 5
B) Find out Z bus matrix if you add Zb from existing bus K to the reference bus/node, assume required data 5

Q.2
A) Find the expression for fault current of n-bus system using Y bus approach and Z bus approach. 6
B) Define power system stability. Write down classification of stability 6

Q.3 Solve any ONE
A) A three bus system is shown in figure the relevant per unit line admittances on 100MVA base are indicated on the diagram and the bus data are given in the table from Y_{bus} and determine voltages at bus 2 and 3 after the second iteration using Gauss-sidle method. Take the acceleration factor α=1.6 12

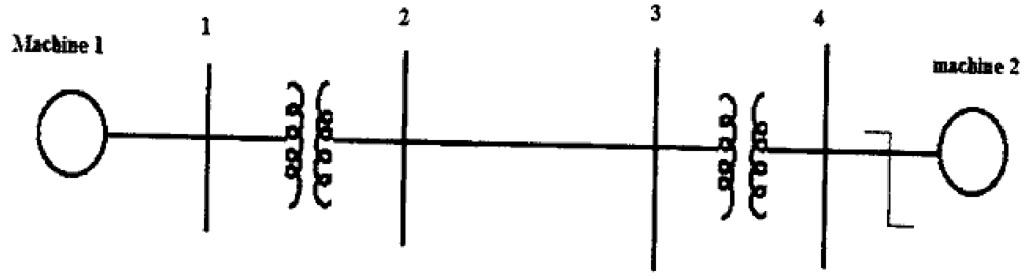


Bus no.	Type	Generation		Load		Bus voltage	
		P_G(MW)	Q_G(MVAR)	P_L(MW)	Q_L(MVAR)	V_{PU}	∂_{deg}
1	slack	?	?	0	0	1.02	0^0
2	PQ	25	15	50	25	?	?
3	PQ	0	0	60	30	?	?

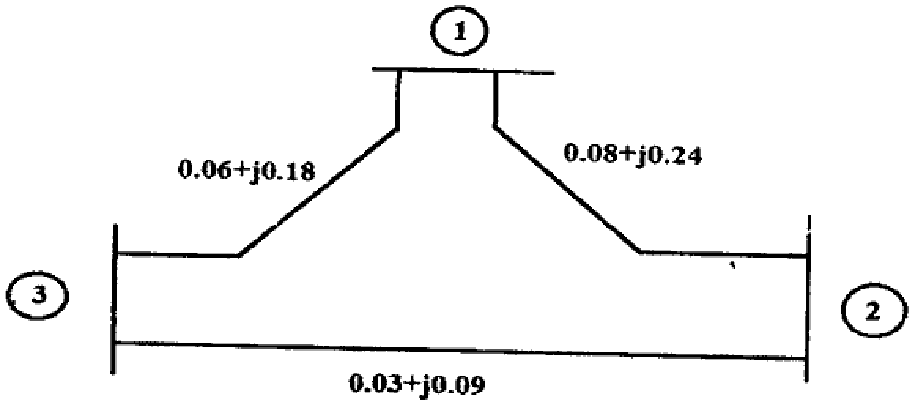
B) Write static load flow equations and derive Gauss-seidel method using Y_{bus} for load bus and PQ bus. 12

Q.4 Solve any TWO

A) A synchronous generator and a synchronous motor each rated 50MVA, 11KV having 12% sub-transient reactance are connected through transformer and a line as shown in figure bellow the transformer are rated 50 MVA, 11/132 KV and 132/11 KV with leakage reactance of 8% each the line has a reactance of 15% on a base of 50 MVA , 132 KV the motor is drawing 25 MW at 0.8 PF leading and a terminal voltage of 10.6 KV when a symmetrical three phase fault occurs at the motor terminals, find the sub-transient current in the generator, motor and fault. 6



B) Determine Y_{bus} for a 3 bus system shown in fig. and neglect the shunt admittances of the line 6



C) Derive Swing equation for transient stability study. 6

Q.5 Solve any TWO

A) Write down Z bus matrix formation when we add Zb in between two existing bus J & K 6

B) Find out admittance matrix with 3 bus and 3 line system and write down characterises of Ybus matrix 6

C) Write down comparison of newton Raphson and gauss siedel method 6

Q.6 Solve any TWO

A) Write down power flow solution using Newton Raphson method. 6

B) Explain equal area criterion for stability study of power system 6

C) Write down power flow solution using fast decoupled load flow 6

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